

Crop Stress Detection with Multispectral Imaging using IA

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Abstract—The article addresses the problem of crop production in the face of biotic and abiotic factors. Highlights the importance of integrated management strategies to optimize resources and increase crop resilience. Multispectral images captured by an Unmanned Aerial Vehicle (UAV) are used to detect stress in crops. The article presents the *K-means* segmentation method and highlights its superiority over the *Otsu* threshold method for efficient segmentation. The relevance of vegetation indices and machine learning in stress assessment is emphasized. The proposed methodology includes data preprocessing, crop detection and stress identification, integrating various techniques. Experimental results demonstrate the effectiveness of this implementation in differentiating healthy and stressed crop regions. Taken together, the study contributes to precision agriculture by combining advanced technologies and innovative techniques for more accurate stress detection and more efficient crop management.

Index Terms—Stress detection, segmentation, multispectral imaging, remote sensing, satellite networks

I. INTRODUCTION

Crops are essential for food security and economic development in many countries. However, various biotic and abiotic factors threaten its production, which can cause plant stress and negatively affect its growth, development, and yield [1]. Among these factors are water deficit, pests, diseases, weeds, salinity, temperature, and radiation, which can reduce the quality and quantity of agricultural products [2]. To face these challenges is necessary to implement integrated management strategies that allow optimizing the use of resources, improving the efficiency and sustainability of production, and increasing the resilience of crops [3]–[5]. One of these strategies is the early and accurate detection of stress in crops, which makes it possible to identify problems, diagnose the causes, assess the impact and apply corrective measures [6]. Detecting stress in crops requires the use of advanced technologies that provide detailed and precise information on the state of plants at a spatial and temporal level. Various works have

demonstrated the effectiveness of spectral data processing, computer vision, and machine learning for this purpose [7]–[10]. These techniques make it possible to extract relevant information from images captured by various sensors, such as RGB, hyperspectral, multispectral, or thermal cameras, and classify plants according to their health or stress [11]–[13]. In [7], a method based on self-organizing hierarchical classifiers was proposed to detect biotic and abiotic stress in wheat and barley crops using hyperspectral imaging. The applications of machine learning for stress phenotyping in plants using different data types, such as spectral, morphological, thermal, or fluorescence, were reviewed in [8]. In [9], the use of deep learning for stress imaging in plants was discussed, highlighting its advantages, challenges, and possible future directions. Finally, in [10], an overview of the trends and prospects of computer vision technology for stress and disease detection in precision agriculture was given, covering different types of crops, leaves, fruits, and vegetables.

Multispectral images are those that capture information in an interval that comprises from 3 to 15 spectral bands, with an approximate width of 100 nm. They are a powerful tool to obtain a more complete and detailed vision, and they capture data within wavelength ranges across the electromagnetic spectrum. These ranges can include visible light and other frequencies such as infrared, revealing additional information about its physical and chemical properties [14].

From these data, it is possible to extract characteristics such as vegetation indices that have shown a high correlation with the physiological and sanitary state of the plants [7]. Each of these indices has its advantages and limitations, depending on the type of crop, plant cover, soil moisture, atmosphere, and the sensor used.

The different spectral resolutions of the sensors that obtain these images could significantly affect these vegetation indices. The remote sensing platform using Unmanned Aerial Vehicles

(UAVs) provides an inexpensive and easy way to obtain high-resolution images [15].

Computer vision is a branch of artificial intelligence responsible for processing and analyzing digital images to extract useful information. In the agricultural context, it constitutes a valuable tool for identifying, analyzing, and diagnosing the different elements present in crop images [10]. Machine learning, for its part, allows us to extract information and knowledge from large volumes of data using algorithms that learn automatically [16].

In addition, in order to detect stress in crops, it is necessary to identify the area occupied by plants in multispectral images previously. This could be achieved using segmentation techniques, which consist of dividing images into homogeneous regions based on criteria such as color, texture, or shape [15]. Segmentation allows separating the crop from the background (soil, water, clouds). This facilitates the calculation of indicators such as vegetation cover or potential yield [17].

In the previous work [18], we proposed an analytical method to determine stress in sugarcane crops from multispectral images. One of the most critical phases of this study was the detection of the crop of interest, for which a segmentation method based on the threshold was used, which, although it obtained acceptable results, could be verified that it was not the best. In this work, we propose to evaluate and compare the efficiency of detecting the crop of interest and its relationship with the detection of stress.

Section II presents the description of the Satellite Internet System for the Monitoring and Analysis of Crops. Section III shows the methodology to detect stress in crops. Section IV describes sugarcane cultivation's design and experimental results. We conclude and analyze the limitations and future work in section V.

II. SYSTEM DESCRIPTION

Figure 1 describes a satellite Internet system for crop monitoring and analysis that uses advanced technologies for precision agriculture. The system includes the components required for its implementation.

The Farm requires a multispectral camera, a georeferencing

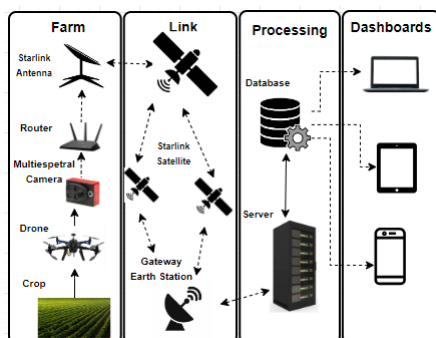


Fig. 1. Satellite Internet System for Crop Monitoring and Analysis

device, an Drone, a router, and a Starlink antenna. The multispectral camera is a camera that captures images at different

wavelengths of the electromagnetic spectrum and is used to detect the state of health, water stress, biomass, or chlorophyll in plants. The drone flies over the crop and takes images with a multispectral camera, labeling them with positioning information and the flight plan. These data are sent over the internet using a router that is the one that distributes the signal to the other devices on the farm. The Starlink antenna connects to the nearest Starlink satellite and provides internet by sending and receiving data via radio waves.

The Link section is the means of transporting data between the farm and the data center. It includes the Starlink satellite network and the Gateway Earth Station. Starlink satellites are small satellites that orbit the Earth at an altitude of about 550 km and that communicate with each other using optical lasers. Each satellite has four dish-shaped antennas that allow it to dynamically point at users and other satellites. Starlink satellites form a mesh network that guarantees global coverage and low latency. The Starlink ground station is a facility that is responsible for sending and receiving data from Starlink satellites and connecting them to the terrestrial Internet network.

The Processing section represents the data center where the information received is processed and made available to end users. It includes a processing server and a database. The data collected by the camera and the processed data are stored in the database. The server processes the data stored in the database and applies artificial intelligence algorithms to extract useful information about the crop.

The Dashboards section represents the devices customers use to monitor the farm using pre-configured panels that connect to the database via the internet. The panels are graphical interfaces that show the information processed by the server about the crop, such as maps, graphs, alerts, or recommendations.

III. METHODOLOGY

Figure 2 shows the flowchart for detecting crop stress using multispectral images. The process begins with obtaining multispectral images produced by an unmanned aerial vehicle on the crop under analysis under favorable atmospheric conditions. Then comes a pre-processing stage in which vegetation indices are aligned, corrected and calculated for subsequent analysis. In the next stage, the objective is to detect or separate the crop from the background, for which different techniques such as Thresholding and Clustering can be used. Finally, the stress is detected with the calculated parameters.

A. Data Pre-Processing

The images obtained by the Unmanned Aerial Vehicle (UAV) present geometric errors due to perspective, camera tilt, lens distortion, drone movement, height variation, insufficient overlap, etc. These errors affect the quality and accuracy of image analysis. To solve these problems, it is necessary to create an orthomosaic, a continuous, homogeneous, and rectified image of the crop. An orthomosaic of each band must be constructed [6].

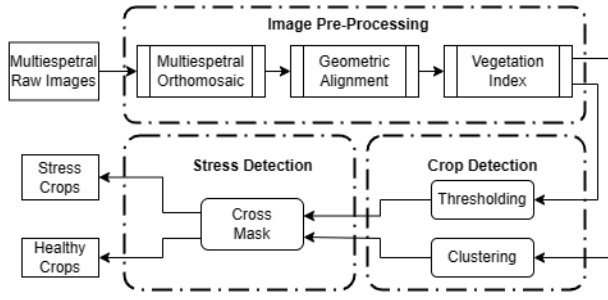


Fig. 2. Flowchart to detect stress in crops

To perform calculation work with several orthomosaics, aligning them and ensuring they have identical dimensions is crucial. Agisoft Metashape helps correct and combine aerial images in the same scale and orientation. In addition, exporting the selected polygon of each orthomosaic allow extract the image and obtaining an exact representation of the crop area of interest. This facilitates the analysis, management, and detection of crop problems in a more precise way [19].

The Vegetation Index (VI) is a parameter that is calculated from the reflectance of light at different wavelengths, which varies according to the amount and state of the vegetation. The vegetation indices are obtained through mathematics between the spectral bands and are useful to differentiate the images according to the physiological and growth parameters of the plants. There are several types of vegetation indices, such as the Normalized Difference Vegetation Index (NDVI), the Soil Adjusted Vegetation Index (SAVI), the Enhanced Vegetation Index (EVI2), and the Normalized Difference Green Vegetation Index (GNDVI), with NDVI being the most widely used and well-known. These indices help detect crop stress due to lack or excess water, nutrients, or pests [20]. Table I shows the vegetation indices and their respective formulas using the bands NEAR INFRARED (NIR), RED, GREEN, and RED EDGE (RE).

TABLE I
VEGETATION INDEX AND THEIR FORMULAS

Vegetation index	Formulas
NDVI	$(\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$
SAVI	$((\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED} + L)) * (1 + L)$
EVI2	$2.5 * ((\text{NIR} - \text{RED}) / ((\text{NIR} + (2.4 * \text{RED}) + 1)))$
GNDVI	$(\text{NIR} - \text{GREEN}) / (\text{NIR} + \text{GREEN})$

B. Crop Detection

Image segmentation is fundamental to identifying and isolating the crop of interest. Segmentation can separate the crop from other elements, such as soil, weeds, or other objects in the image. This can be achieved using different algorithms and techniques, such as thresholding, region-based segmentation, or machine learning.

1) *Threshold: Otsu*: The threshold method separates regions of interest using a defined threshold, enhancing analysis

and noise reduction in applications like edge detection and object segmentation. The *Otsu* method optimally selects a threshold by maximizing inter-class variance. It's effective with noise, flat histograms, and inadequate lighting in real-world images. A modified *Otsu* variant is proposed to reduce computation time [21]. The between-class variance is calculated using the formula in (1) [22].

$$\sigma_B^2(t) = \omega_1(t)(\mu_1(t) - \mu_T)^2 + \omega_2(t)(\mu_2(t) - \mu_T)^2, \quad (1)$$

where:

$\omega_1(t)$ y $\omega_2(t)$ are the probabilities of the classes separated by the threshold t .

$\mu_1(t)$ y $\mu_2(t)$ are the means of the classes separated by the threshold t .

$\sigma_B^2(t)$ is the variance between classes for a threshold t .

μ_T is the overall mean of the image.

The optimal threshold t^* is the one that maximizes the variance between classes, in (2).

$$t^* = \max_{1 \leq t \leq L} \sigma_B^2(t), \quad (2)$$

where L is the number of gray levels in the image.

2) *Clustering: K-means*: Clustering is a machine learning technique that consists of grouping objects or data into groups with similar characteristics, known as clusters. The objective of clustering is to identify patterns or perform segmentation tasks in unlabeled data sets, that is, without prior classification. Clustering is part of unsupervised learning, since the number and nature of the clusters are not known a priori [12]. *K-means* is a clustering algorithm that groups a data set into k groups, or clusters, in such a way that data in the same cluster is more similar to each other than data in other clusters. is defined in (3) [23].

$$\min_S \sum_{i=1}^k \sum_{x \in S_i} |x - \mu_i|^2, \quad (3)$$

where:

S is the set of k clusters, $S_1, S_2, \dots, S_k$.

x is a point in the data set.

μ_i is the centroid of the cluster S_i , that is, the average of the points that belong to that cluster.

$\|x - \mu_i\|$ is the Euclidean distance between the point x and the centroid μ_i .

The objective of the clustering method *K-means* is to find the partition S that minimizes the sum of the squared distances from each point to its corresponding centroid, also called inertia or intra-cluster variance.

3) *Metrics to evaluate segmentation*: Among the metrics used to assess the quality of the segmentation's are:

The Sum of Squared Within (SSW) measures the sum of the squared distances between each point and the centroid of the group to which it belongs. The smaller the WSS, the greater the group cohesion. The formula for calculating inertia is

defined in (4) [24].

$$SSW = \sum_{i=1}^k \sum_{x \in C_i} \text{dist}^2(m_i, x), \quad (4)$$

where:

k is the number of clusters.

x is a cluster point.

C_i and m_i is the centroid of the cluster C_i .

Davies_Bouldin Index is a measure of how far apart there is between groups or regions. It is calculated as the average of the measure of similarity between each group and its most similar group, where the similarity is the ratio between the distances within the group and the distances between the groups. The formula to calculate is defined in (5) [25].

$$\text{Davies_Bouldin Index} = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{S_i + S_j}{d_{ij}} \right), \quad (5)$$

where:

k is the number of groups.

S_i is the mean distance within the i -th group.

S_j is the mean distance within the j -th.

d_{ij} is the distance between the centroids of the groups i and j .

The lower the value of Davies_Bouldin Index, the more distant and distinct the points between the groups.

Calinski_Harabasz Index is a measure that combines variation within and between groups or regions. It is calculated as the ratio between the sum of the squares between the groups and the sum of the squares within the groups, multiplied by a factor that depends on the number of points and groups. The formula to calculate is defined in (6) [26].

$$\text{Calinski_Harabasz Index} = \frac{SSB}{(k-1)} \cdot \frac{n-k}{SSW}, \quad (6)$$

where:

k is the number of groups.

n is the number of points.

SSB is the sum of the squares between the groups.

SSW is the sum of the squares within the groups.

The higher the value of Calinski_Harabasz Index, the better the quality of the segmentation, as it indicates that there is little variation within groups and a lot of variation between groups.

C. Stress Detection

A simple method to detect the stressed crop is to apply the logical AND operation to combine the segmentation of the near infrared (NIR) band, which indicates the presence of the crop, with each binary mask obtained by the calculated vegetation index. For example, in [27] they identify a healthy crop for NDVI with values greater than 0.39.

Another way to detect stress is using machine learning. For this, it is necessary to have a reference of labels of stressed or healthy plants, which ideally are obtained and verified by

experts. It is also required to have a series of characteristics such as spectral bands, vegetation indices, and characterization of texture or color. There are different supervised classification algorithms that learn these labels and allow generalization to new cultures [7].

IV. RESULTS

This study utilized 155 multispectral images from the NIR, RED, GREEN, and RED EDGE bands, acquired through the 3DR RTF X8+ UAV and Parrot Sequoia 3.98 mm Multispectral Camera. These images were taken at an average height of 38.3 meters over an experimental sugarcane field at (ETICA) [18]. Agisoft Metashape software was employed to create georeferenced orthomosaics for each band, merging images from diverse angles. This facilitated subsequent analysis of the sugarcane crop, aiding in detecting stress and health. To assess the health status of the vegetation, different indices were calculated using the spectral bands of the multispectral image. These indices are numerical measurements that relate the different bands, these are sensitive to the photosynthetic activity and the chlorophyll content of the plants. The calculated indices were NDVI, SAVI, GNDVI and EVI2, which have different properties and advantages depending on the terrain conditions and vegetation cover.

Figure 3 represents the result of the calculation of the

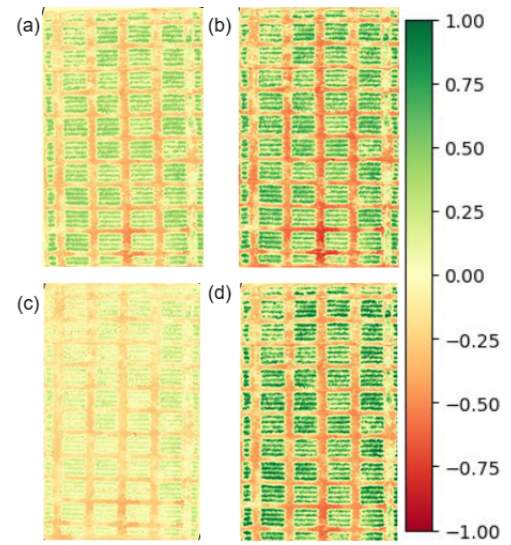


Fig. 3. Results Vegetation Index: (a) NDVI, (b) SAVI, (c) GNDVI and (d) EVI2

vegetation indices. Each graph shows the index value for each pixel of the multispectral image, using a color scale from green (highest value) to red (lowest value). It can be seen that the indices have similar patterns, but with different ranges and contrasts. These graphs make it possible to identify the areas of the crop that present greater or lesser vegetative vigour, as well as the areas affected by water stress or some other factor. Two different methods were used to segment the near-infrared band image: the Threshold: *Otsu* method and the Clustering:

K-means method. Figure 4 shows the segmented images using these two methods. You can see that the *Otsu* method produces a binary image, where the white pixels represent the crop and the black pixels represent the background. The *K-means* method produces an image with multiple levels of gray, with each level representing a different group. The

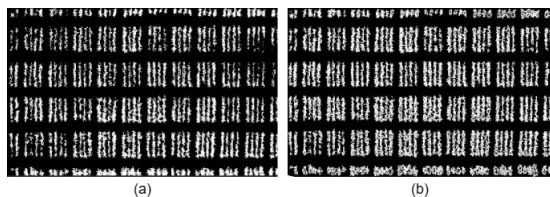


Fig. 4. Segmentation: (a) Threshold: *Otsu* and (b) Clustering: *K-means*

optimal number of clusters for the *K-means* method was $k=2$, because we wanted to segment the image into two groups: the crop of interest and the background. The objective was to differentiate the areas of the image that correspond to the crop from those that correspond to the background. To compare the quality of the segmentation's obtained by each method, three metrics were used: Sum of Squared Within (SSW), Davies_Bouldin Index and Calinski_Harabasz Index. These metrics assess the variation within groups, the separation between groups, and the overall quality of segmentation, respectively. The results showed that both methods produced similar and good quality segmentation's, but with a slight advantage for the *K-means* method in all three metrics. This could be because the *K-means* method fits the data distribution better in this particular case. It is concluded that the *K-means* method is more suitable for the segmentation of this image than the *Otsu* method.

Table II shows the results obtained from the evaluation of the methods using the corresponding metrics.

To identify the stress of the sugarcane crop, the result

TABLE II
SEGMENTATION EVALUATION RESULTS

Method	WSS	Davies-Bouldin	Calinski-Harabasz
<i>K-means</i>	1.65×10^{14}	0.442937	1.3544284×10^6
<i>OTSU</i>	1.65×10^{14}	0.442948	1.3544285×10^6

obtained by calculating the normalized difference vegetation index (NDVI), which is a measure of vegetation health, was used. From this result, two binary masks were created to separate the pixels representing healthy crop and crop stress. Figure 5, you can see these results. Figure 5 (a) shows white pixels representing the healthy crop, while Figure 5 (b) shows white pixels representing crop stress.

To visualize the sugarcane crop's health status, we employed a two-step approach. First, the Clustering method was applied to the near-infrared band image to segment it. This helped isolate crop pixels from background and noise, enhancing accuracy. Concurrently, binary masks were generated through NDVI calculations, categorizing pixels into healthy and stressed classes based on their NDVI values. Merging the segmented image

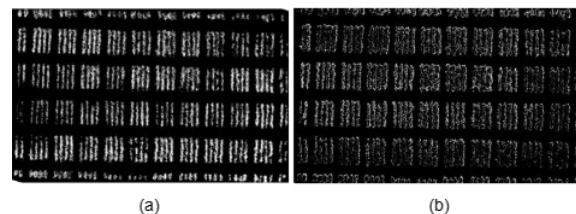


Fig. 5. Binary Mask: (a) Healthy Crop and (b) Stressed Crop

and binary masks using logical AND operations produced a composite image Figure 6. The resulting image presents the crop's health status: green pixels indicate healthy regions, while red pixels signify stressed areas. This composite aids in evaluating stress distribution and guiding intervention strategies for crop management.

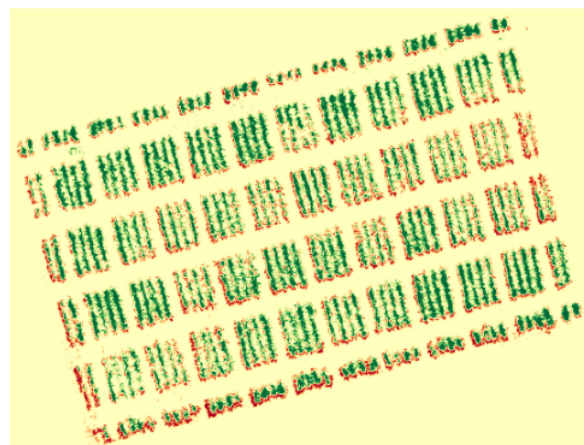


Fig. 6. Sugarcane crop: Healthy: Green and Stressed: Red

V. CONCLUSIONS

In this study, crop stress detection was explored using advanced technologies and innovative approaches. The implementation of a satellite and Internet-based monitoring system made it possible to effectively evaluate the health of the crops. A key contribution was the introduction and evaluation of the *K-means* segmentation method to identify areas of interest in multispectral images. Compared to the threshold method *Otsu*, *K-means* demonstrated more accurate segmentation. Furthermore, the importance of vegetation indices and the integration of machine learning in crop stress assessment was highlighted. This study advances precision agriculture by combining emerging technologies and innovative methods to improve crop stress detection and management.

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